CS 254 Machine Learning
Project Final Report

1. Introduction

We are using machine learning to determine the recipients of the National Football League's most coveted individual awards, given by the Associated Press. The Most Valuable Player (MVP), Offensive Player of The Year (OPOY), Defensive Player of The Year (DPOY), Offensive Rookie of The Year (OROY), and the Defensive Rookie of The Year (DROY). These awards are often very difficult to predict because of many different factors: coaching, scheme, teammates, etc. And because it is difficult, Vegas turns a profit off people who like to waste money gambling on these awards.

Our approach is going to be trying to find the best machine learning model that most accurately predicts winners and non-winners for each award. This is related to other work in the area because sports analytics is exploding right now. Teams and media companies have data analytics departments who want to predict the best of the best each year. Teams will try to predict how likely a catch is to be completed, average yards over expected, probability of scoring, and many other events and outcomes. At the end of our experimentation, we aim to have a model that can somewhat accurately predict the winner of each award above via vote count and gain insight on what features are most important in these votes and how they are a reflection of biases and media narratives.

After our experimentation, our models approaching this problem in a regression context performed adequately with R^2 values between 0.19 and 0.43. However, our classification models were considerably powerful with accuracies between 0.84 and 0.96 and ROC AUC scores between 0.88 and 0.99. Our models show that media voting is very strongly influenced by the statistics, but the non-quantifiable aspects of football are still relevant in the judgement of these panelists.

2. Problem Definition and Algorithm

2.1 Task Definition

The problem we are looking to answer is how much each statistic contributes to the number of votes an NFL player receives for an award. The statistics we are interested in are general player statistics, like games played and years of experience, as well as position-specific statistics, such as pass completion rate for quarterbacks and rushing yards per game for running backs. The models we create for each award will be trained on these statistics and predict the number of votes a given player will receive in some award. The accuracy of our models will be

evaluated more heavily on if it can correctly predict the award winner, the player with the most votes, rather than the number of votes itself. This problem is very interesting because awards build a player's legacy and are viewed as proof of greatness. However, it is very unlikely to attain unanimous agreement for which player was the greatest in any given season due to biases and media narratives. The most objective truth one can rely on for these debates are statistics. We hope that the results of our analysis will show how much statistics contribute to a player's case for an award and how the differences between our expectations and reality reflect on inherent biases in football.

2.2 Algorithm Definition

The primary algorithm we expect to use for most awards is a random forest algorithm to predict the number of votes a player will receive after a season. We chose the random forest because it evaluates feature importance, which is crucial for our objective of determining what statistics are considered significantly in award voting. Furthermore, the random forest is generally high-accuracy and low-variance, which will help in this context because the statistics of award winners vary considerably between seasons. We will also experiment with other algorithms such as linear regression, logistic regression, gradient boosting, XGBoost, and neural networks. Our neural networks are SKLearn's multi-layer perceptrons regressor and classifier models with default settings, that is with hidden layer shapes of (100,), ReLU activation, Adam solver, and squared loss error. We will use the models in both regression and classification approaches to compare which is more appropriate.

3. Experimental Evaluation

3.1 Methodology

We hypothesize that machine learning models are powerful tools in predicting NFL award winners, more specifically that there exists at least one statistically significant variable that is a strong indicator of a player's vote total. We will evaluate our methods for its accuracy in predicting the number of votes a player has received if the model is a regressor or by its ability to correctly identify winners and non-winners if the model is a classifier. As mentioned above, we expect the algorithm to perform better as a regressor than a classifier due to the class imbalance between award winners and non-award winners. When creating the models, we randomly partitioned 20% of the dataset for testing and the remainder for training each award's model. Each random forest model was tuned to one hundred estimators, maximum tree depth of ten, and set to the same random state. Then, we examined each model's performance with the test set with R^2 scores. If the R^2 score for a model is close to one, then it is a helpful predictor in the award voting. If it is closer to zero, then it is a poor model. As explained above, we proceeded with the random forest as the baseline given its characteristic of high accuracy and low variance, as well as its informative feature importance output. With these models, we are testing our hypothesis

that data from past seasons can be used to accurately anticipate future award winners in the NFL MVP, OPOY, DPOY, OROY, and DROY categories.

3.2 Results

3.2.1 Regressor models

MVP:

Best Model: Random Forest & Neural Network Ensemble

Performance Metrics:

R^2 Score: 0.4344

RMSE: 12.5392

MAE: 9.4485

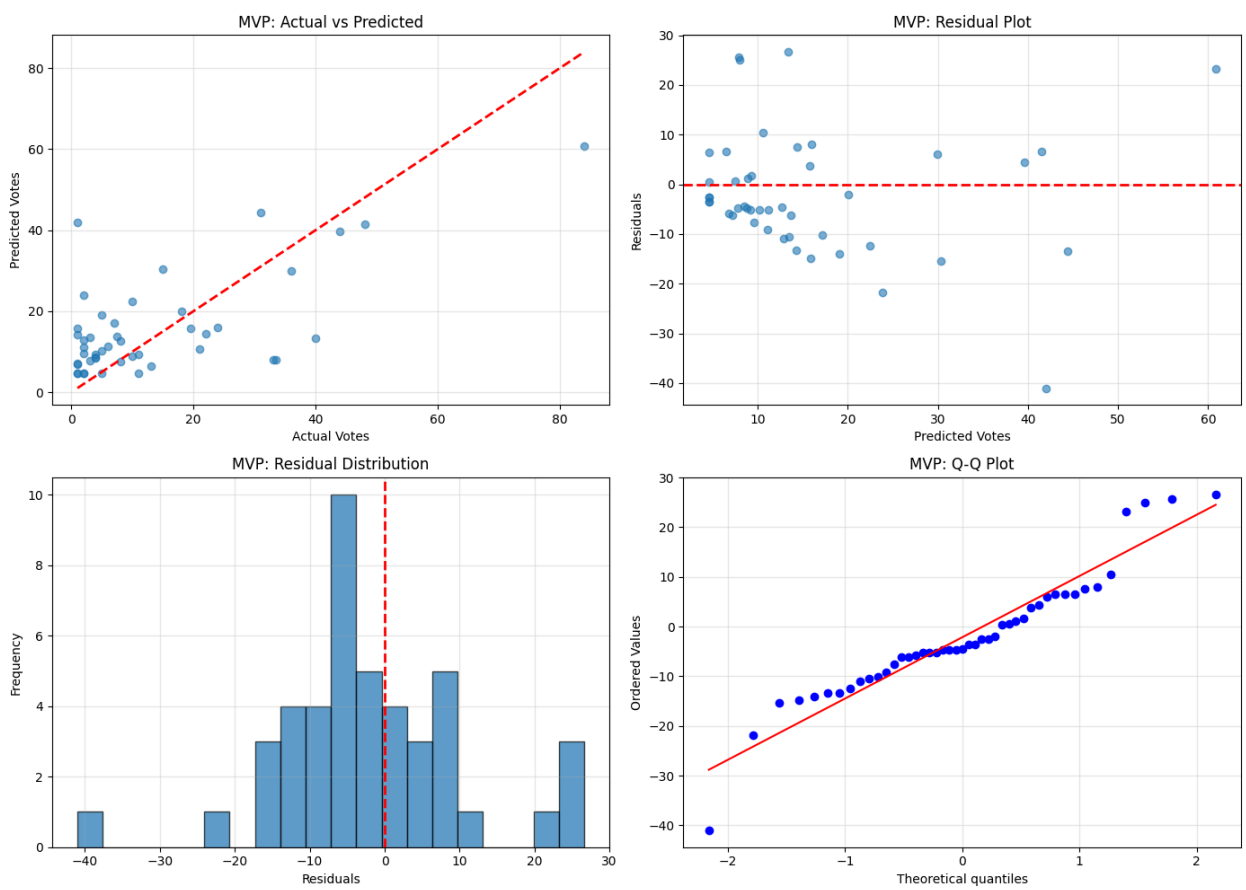
Residual Analysis:

Mean residual: -2.1601 (ideally ~0)

Std dev: 12.3517

Min residual: -41.0041

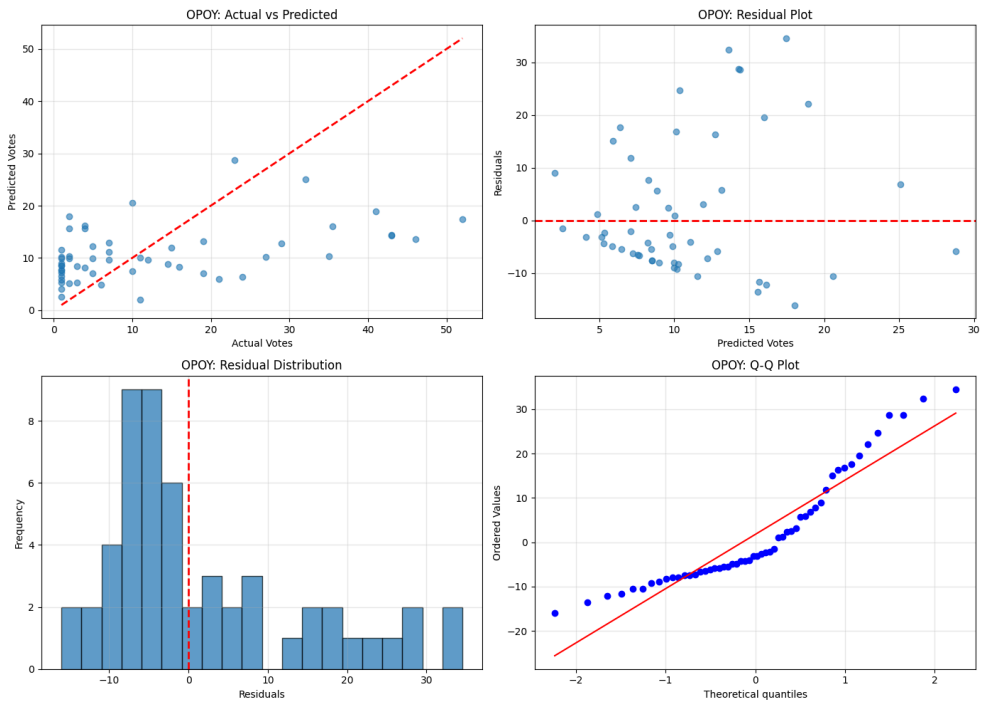
Max residual: 26.6499



OPOY:

Best Model: Random Forest
Performance Metrics:
  R^2 Score: 0.1911
  RMSE: 12.7158
  MAE: 9.8507

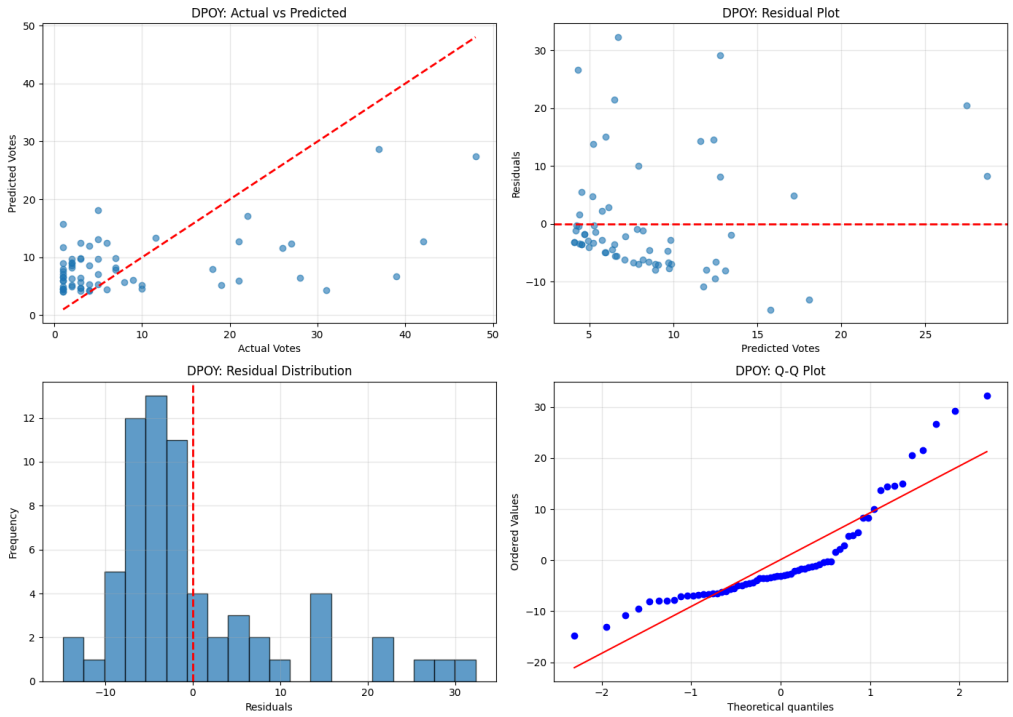
Residual Analysis:
  Mean residual: 1.7599 (ideally ~0)
  Std dev: 12.5934
  Min residual: -16.0104
  Max residual: 34.5409



DPOY:

Best Model: Random Forest
Performance Metrics:
  R^2 Score: 0.2480
  RMSE: 9.8170
  MAE: 7.1760

Residual Analysis:
  Mean residual: 0.1021 (ideally ~0)
  Std dev: 9.8165
  Min residual: -14.7872
  Max residual: 32.3098



OROY:

Best Model: Gradient Boosting

Performance Metrics:

R^2 Score: 0.3013

RMSE: 20.0640

MAE: 15.0490

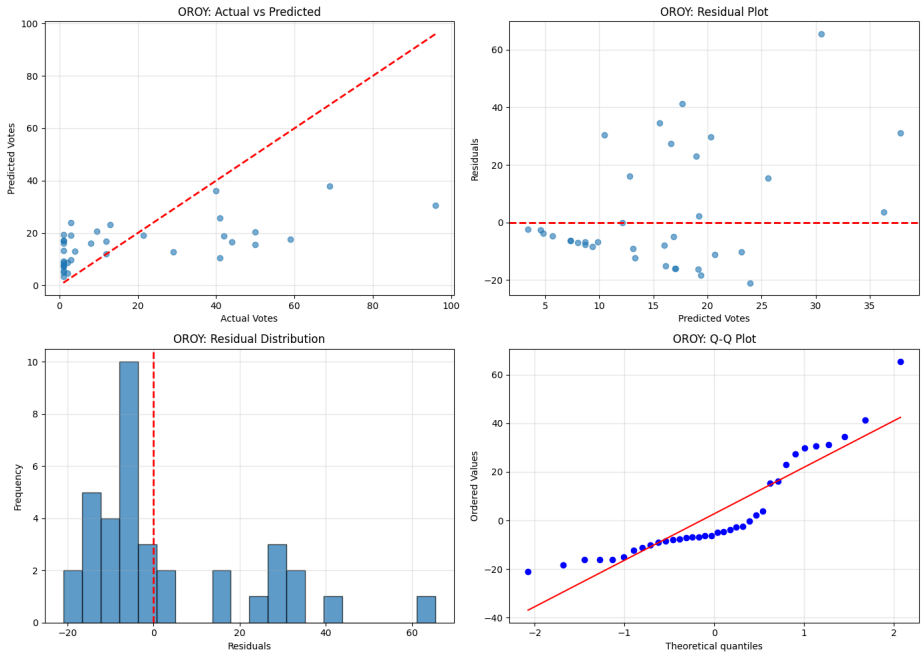
Residual Analysis:

Mean residual: 2.7669 (ideally ~0)

Std dev: 19.8723

Min residual: -20.9441

Max residual: 65.4726



DROY:

We were unable to get a good linear DROY prediction, so decided to switch to classification for this award, simply predicting whether or not a player won the award.

Best Model: XGBoost

Performance Metrics:

R^2 Score: -0.0907

RMSE: 19.5112

MAE: 14.4445

Residual Analysis:

Mean residual: 2.1450 (ideally ~0)

Std dev: 19.3930

Min residual: -28.0751

Max residual: 49.0388

DROY Classification model (Winner vs Non-Winner)

	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.727273	0.384615	0.555556	0.454545	0.780952
Random Forest	0.818182	0.666667	0.222222	0.333333	0.704762
Gradient Boosting	0.818182	0.600000	0.333333	0.428571	0.707937
XGBoost	0.795455	0.500000	0.444444	0.470588	0.666667

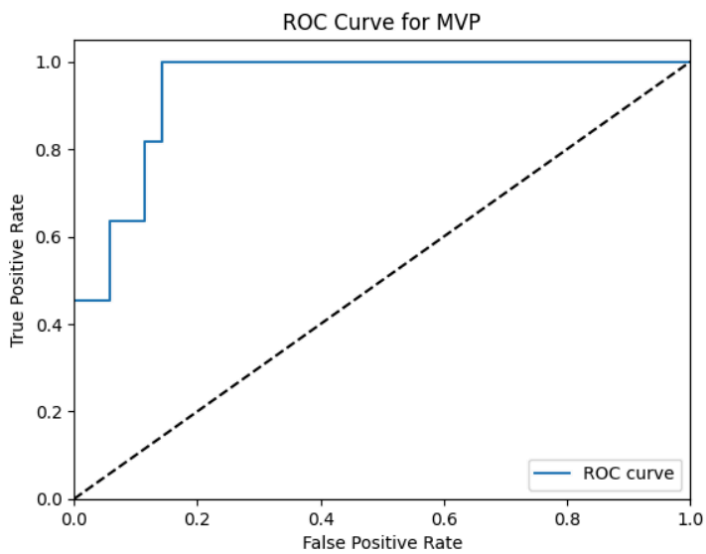
3.2.2 Classification Models

MVP:

Best Model: XGBoost

Performance Metrics:

Test accuracy: 0.8478
Precision: 0.6667
Recall: 0.7273
F1: 0.6957
ROC AUC: 0.9429

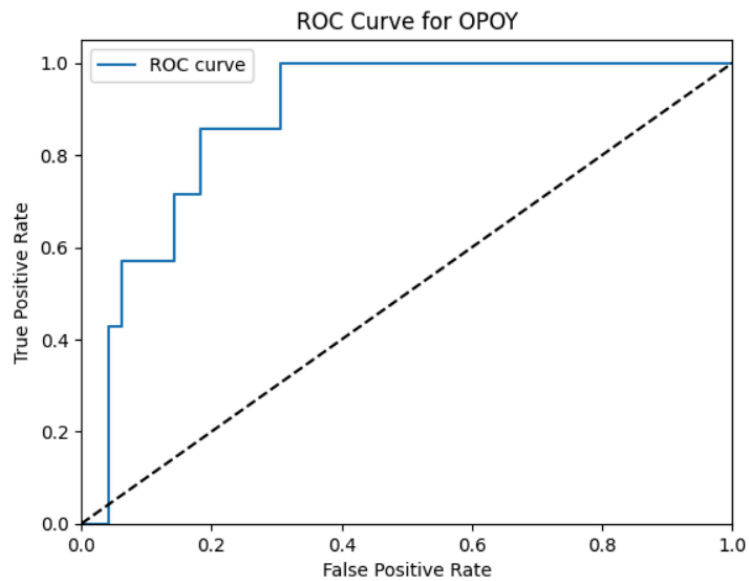


OPOY:

Best Model: Random Forest

Performance Metrics:

Test accuracy: 0.8214
Precision: 0.3636
Recall: 0.5714
F1: 0.4444
ROC AUC: 0.8834



DPOY:

Best Model: Gradient Boosting

Performance Metrics:

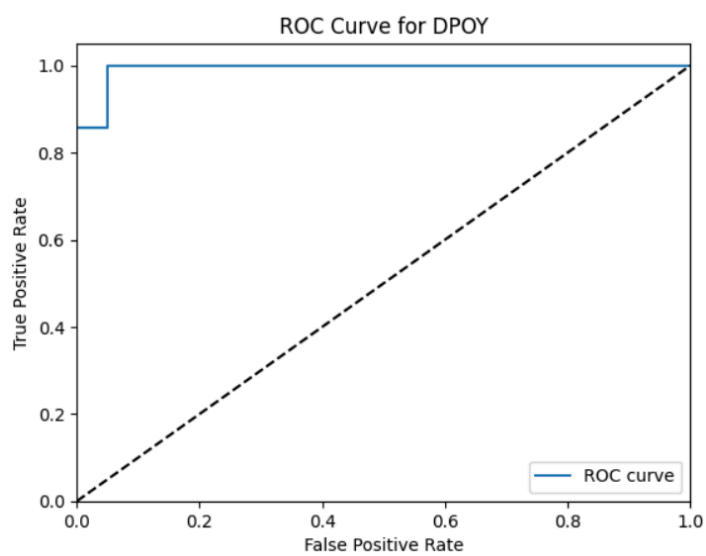
Test accuracy: 0.9552

Precision: 0.7500

Recall: 0.8571

F1: 0.8000

ROC AUC: 0.9929

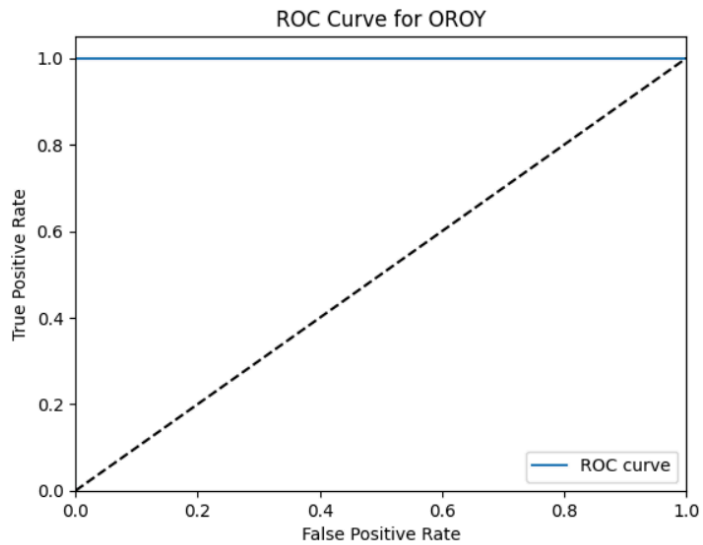


OROY:

Best Model: Random Forest

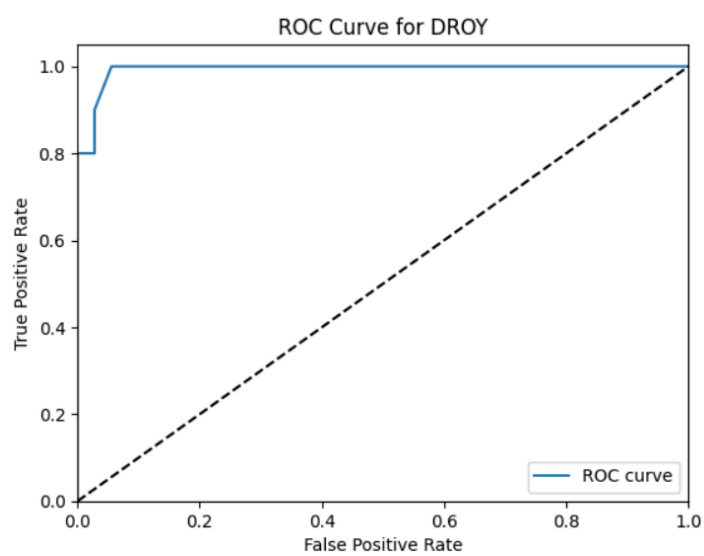
Performance Metrics:

Test accuracy: 0.9459
Precision: 0.7778
Recall: 1.0000
F1: 0.8750
ROC AUC: 0.9857



DROY:

Best Model: XGBoost
Performance Metrics:
Test accuracy: 0.9438
Precision: 0.7692
Recall: 1.0000
F1: 0.8696
ROC AUC: 0.9931



* See GitHub for confusion matrices

3.3 Discussion

The results partially support the original hypothesis: machine learning models can capture meaningful patterns in NFL award voting, but their predictive accuracy is limited by the structure and variability of the data. Across the MVP, OPOY, DPOY, and OROY awards, the ensemble regression models produced moderate R^2 values (0.19 - 0.43), indicating that a non-trivial portion of variance in vote totals can be explained by player statistics alone. Meanwhile, the classifier models produced very strong test accuracy (0.84 - 0.96), demonstrating that the player statistics are quite reliable in discerning between a winner and a non-winner. However, there still exists a small amount of unexplained variation which reflects the fact that award voting is influenced by contextual, narrative, and team-success factors that are difficult to encode numerically.

4. Related Work

In Watters' analysis, they use machine learning to calculate the probability of one team beating another. They pulled comprehensive play-by-play data from each season between 1999 and 2021. From that data, they primarily selected offensive features like third down conversion rate and scoring drives, as well as off-field factors like home field advantage and team elo rating. Then, they trained and evaluated a variety of models such as logistic regression, SVM, boosted tree, and nearest neighbors which receives these inputs and outputs a value between 0 and 1 representing the home team's probability of defeating the away team for every game in a season. Our problem approach is different in that it will heavily rely on an individual player's performance rather than a team's performance. We are using aggregated data amassed across the whole NFL regular season, while Watters is using detailed play-by-play data that is most relevant within each game or between games. Furthermore, Watters omits defense data while it is crucial for our research as it will be required for predicting MVP, DPOY, and DROY.

5. Code and Dataset

We sourced our data from the publicly available website Pro-Football Reference. The data includes award voting results in each of the five award categories from 1980 to present as well as the statistics of each award candidate. The data is already well-labeled and contains many features, however it requires some cleaning as each season is on a separate page. Additionally, it is slightly lacking in some defensive statistics, so models predicting defensive award winners will be short-sighted against players excelling in coverages. We also added some interaction features such as pass touchdown rate and sack conversion rate which are a better measure of player quality than the statistics' individual counterparts. The cleaned data files for each award vote can be found in the project's GitHub.

6. Conclusion

This project demonstrated that machine learning methods can capture meaningful patterns in NFL award voting, although the degree of predictability varies significantly by award. Using a full pipeline (including data cleaning, feature engineering, model tuning, and ensemble construction) we produced competitive models for MVP, OPOY, DPOY, and OROY, achieving test R^2 values between 0.19 and 0.43 in the case of regressors, as well as test accuracies between 0.84 and 0.96 through classifiers. These results indicate that a substantial portion of year-to-year variation in voting can be explained by player statistics alone. For regression models, MVP was the most predictable award, reflecting the strong alignment between quarterback efficiency metrics and voting behavior. OPOY, DPOY, and OROY displayed weaker but still meaningful predictability. For classification models, all models produced strong metrics with the most accurate predictions being DPOY's model. For both contexts, the models are not infallible due in part to the wide variation in positional roles and the influence of contextual, narrative, and team success factors not easily captured numerically.

A key contribution of this work was the use of ensemble models, which consistently outperformed the best standalone algorithms. By combining Random Forests, Neural Networks, and XGBoost, the ensembles were able to model nonlinear relationships, feature interactions, and threshold effects inherent in football performance data. This highlights the value of blending complementary model types when modeling complex human-driven outcomes such as award voting.

An important insight emerged in analyzing Defensive Rookie of the Year (DROY). Due to extremely sparse vote distributions and limited sample size, regression approaches failed to outperform a naive baseline. Reframing the task as a binary classification problem (predicting the winner versus non-winner) led to dramatically better performance, with models achieving 80%+ accuracy and strong ROC-AUC values (0.66–0.78). This shows that ML success depends not only on choosing the right model, but on appropriately defining the prediction task itself. In this case, classification aligned better with the structure of the label and produced clear, interpretable predictions.

With more time, we would like to tune and improve the regression models to be better predictors of votes or vote share. Additionally, we would like to make predictions for the 2025 award winners as well as expand into the Hall of Fame voting and create machine learning models that predict whether a player's career is legendary enough to receive the highest title in all sports. The results of our models could be used in sports betting, either to create odds determined by the probabilities outputted from our models or to exploit market inefficiencies and turn profits.

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